

ANIKET PATIL

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SUMMARY

Machine Learning Engineer with expertise in deep learning, computer vision, and production AI deployment. Built a RAG pipeline with sub-2s latency and 94% faithfulness, and developed a semi-supervised cardiac segmentation model at Indian Institute of Technology Ropar that outperformed fully supervised approaches using only 20% labeled data. Published 2 Springer papers and earned Kaggle Master recognition.

EDUCATION

Vishwakarma Institute of Information Technology, Pune Jun 2021 – Jun 2025
B.Tech in Artificial Intelligence and Data Science | GPA: 8.88/10.0 Pune, India

EXPERIENCE

Research Intern | Indian Institute of Technology, Ropar Jun 2024 – Feb 2025

Dynamic Uncertainty Correction for Semi-Supervised Medical Image Segmentation

- Architected an **uncertainty-guided semi-supervised segmentation framework** for cardiac MRI using MC Dropout (mean-variance) to dynamically correct pseudo-labels under limited annotation budgets, advancing clinical AI deployability.
- Achieved **>50% boundary error reduction** using only 20% labeled data on the ACDC dataset (HD95: 10.62 → 4.85; ASD: 3.04 → 1.49), surpassing fully-supervised baselines.
- Engineered an **end-to-end PyTorch pipeline** (preprocessing → training → evaluation → visualization) with Git version control, reducing model development cycles by **40%**.
- Delivered a **1.8M-parameter model** that outperformed established semi-supervised baselines, demonstrating high parameter efficiency suitable for edge and clinical deployment.

Research Intern | Vishwakarma Institute of Information Technology, Pune May 2023 – Aug 2023

Lightweight SegNet-Based Framework for Dermoscopic Lesion Segmentation

- Designed a **modified SegNet encoder-decoder CNN** for automated skin lesion segmentation on dermoscopic images using computer vision techniques, reducing learnable parameters while preserving fine-grained spatial detail.
- Outperformed baseline models by **+7.38 pp Jaccard** and **+5.30 pp Dice** on the ISIC 2017 dataset (2,000 train / 600 test images), demonstrating robust generalization.
- Optimized architecture depth for **efficient real-world inference**, suitable for resource-constrained clinical and edge environments; findings contributed to two peer-reviewed publications.

PROJECTS

VisualSentry — Industrial Visual Anomaly Detection System Apr 2026 – Jul 2026

- Designed and built **VisualSentry**, a production-grade **industrial anomaly detection** system fusing **PatchCore (DINOv2 + FAISS)** and **WinCLIP (CLIP zero-shot learning)**, trained only on normal images with no labeled defect data required.
- Diagnosed and fixed a prototype collapse bug in zero-shot CLIP scoring by redesigning it into **per-prompt similarity banks with top-k aggregation**, restoring the model's discriminative power.
- Achieved **0.979 Image AUROC** and **0.954 Pixel AUROC** on the MVTec AD benchmark through weighted score fusion, and built an **ONNX deployment** pipeline enabling **sub-50ms GPU inference** for production use.

Realtime Monocular Depth Mapping | [GitHub](#) Sep 2023 – Jan 2024

- Designed a **U-Net encoder-decoder CNN** for dense depth map generation from single RGB images; achieved real-time inference at **35.5 FPS (6× speed, 14× fewer parameters)** vs. baseline using OpenCV for preprocessing.
- Integrated composite loss (L1 + SSIM + gradient) for enhanced edge preservation, achieving **MSE 0.04588** and **SSIM 0.7779** on NYU Depth V2, outperforming standard approaches.

Portfolio Optimization Using Market Correlations | [GitHub](#) Oct 2022 – Feb 2023

- Developed and benchmarked ARIMA, LSTM, and Prophet time-series models on NYSE, NASDAQ, and S&P 500 data; identified LSTM as top performer via MSE/RMSE with rolling cross-validation.
- Conducted sectoral correlation analysis using Pandas and Matplotlib to derive data-driven investment insights for portfolio optimization.

PUBLICATIONS

- Enhancing Skin Lesion Diagnosis with Data Augmentation Techniques: A Review | [Paper](#)**
Aniket Y. Patil, Anjula Mehto, Saif Nalband | Multimedia Tools and Applications, Springer, 2024
- Skin Cancer Image Augmentation Techniques Using AI: A Survey | [Paper](#)**
Aniket Y. Patil, Yashwant S. Ingle, Nuzhat Faiz Shaikh and Parikshit Mahalle | Springer, 2023

SKILLS

ML / DL Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn
Generative AI & Automation:	LangChain, RAG, HuggingFace Transformers, OpenAI API, n8n
Data & Analytics:	NumPy, Pandas, SQL, Matplotlib, Tableau, PowerBI
Languages:	Python, R, LaTeX
MLOps & Deployment:	Git/GitHub, Linux, Google Colab, AWS, Jupyter, Docker, FastAPI

ACHIEVEMENTS

- Kaggle Master — [Profile](#)**: Notebooks awarded bronze medals; recognized for high-quality exploratory data analysis and algorithm tutorials across multiple public datasets.